Standby User Guide

Revision

Version	Date	Author/Revision	Author/Revision History
1.2.0	2023/09/12	Jerry.wang	

Overview

Standby driver is mainly used for low-power standby during the bootloader phase and main code phase, and wake-up operations.

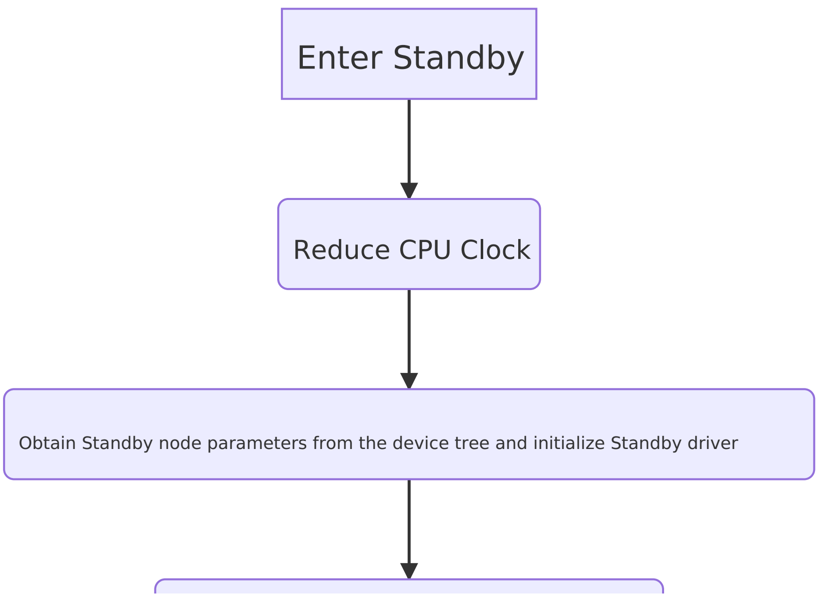
Note: The current Standby wake-up mechanism used is CPU reset.

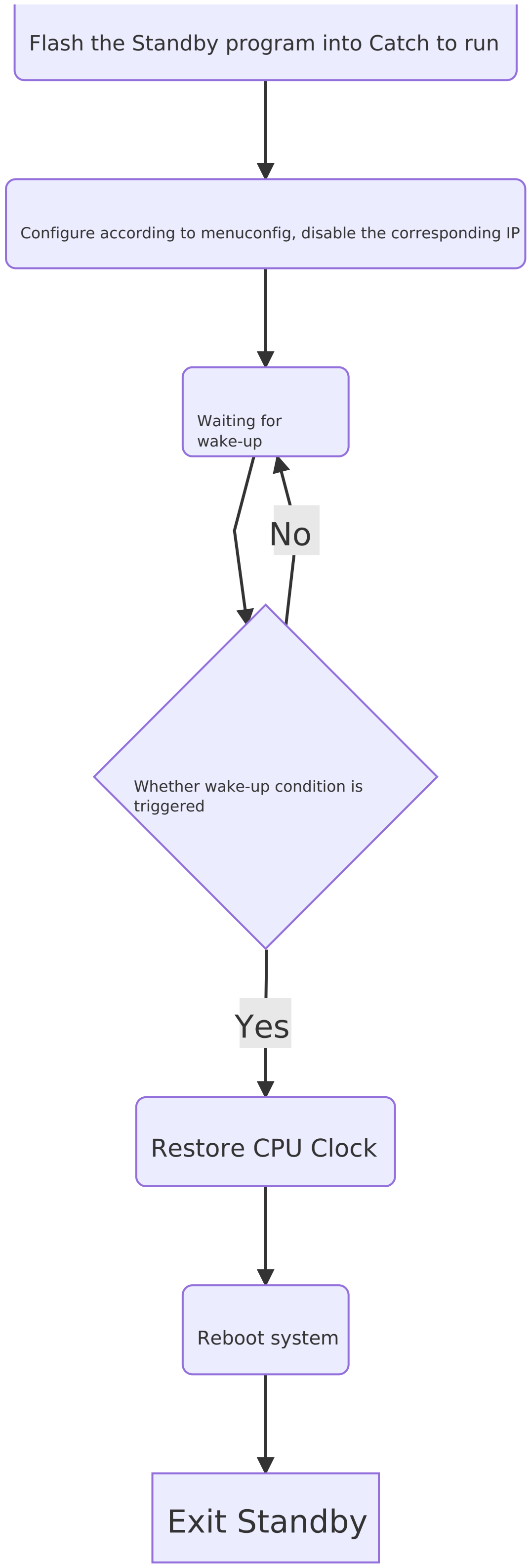
Audience

This document (this guide) is mainly applicable to the following engineers: Technical Support Engineer Software Development Engineer

Module

- Overall Framework





- Directory structure
    - Source code:

```
hcrtos/components/kernel/source/drivers/standby/
```

- DTS configuration

```
standby {
        ir = <28 0>;                // <powerkey_scancode0, powerkey_scancode1> Allow setting multiple
power key
```

```
        adc = <1 0 850>; // <ADC channel minimum wake voltage value maximum wake voltage value> ** Note: adc key wake-up
Unless there are special circumstances, please make sure to use the button with a voltage value of 0 as the adc power key
        gpio = <PINPAD_L0 0>                // <PIN used to wake up with gpio button, polarity at wake-up>
        ddr-gpio = <1 0>; // <PIN that controls DDR power-off, polarity when powered off>
};

If infrared and ADC key wake-up are needed, enable the irc and key-adc@x nodes (i.e., configure the node parameters, and also set the
status to "okay")

irc {
        pinmux-active = <PINPAD_XXX 1>;
        linux,rc-map-name = "xxx-xxx";
        status = "okay";
};

key-adc@1 { // standby uses the adc channel, which should be the same as this node
        status = "okay";
        adc_ref_voltage = <XXX>; // range: 0-2000mV
        key-num = <xxx>;                    //demo---map2--projector
        key-map = <xxx xxx xxx>,//            up 0.41 ->left
                <xxx xxx xxx>,
                <xxx xxx xxx>,
                <xxx xxx xxx>,
                <xxx xxx xxx>,
};

When using the standby driver, infrared and ADC wake-up must be used, the ***irc node and key-adc node*** must be enabled
*** Note: In Linux standby, you also need to enable the standby node, the irc node, the key-
adc node ***
```

- menuconfig configuration
    - Driver selection

```
 > Components
      > kernel
            > Drivers
                  [*] standby driver            --->
```

    - Wake Up Mode selection

      Currently supports infrared, ADC, and GPIO - three wake-up modes for Standby.

```
> Components > kernel > Drivers > standby driver > Wake Up Mode
      --- Wake Up Mode
      [*]       ir wake up
      [*]       adc wake up
      [ ]       gpio wake up
```

    - close ip: select through menuconfig which ip to close; typically requires changes. If standby requires controlling the high/low level of pins T0~T5, do not select [close sdio] and [close lvds]
      [close sdio]

```
> Components > kernel > Drivers > standby driver
--- standby driver
Mode (Enable close ip)              --->
[*]       close sdio
[*]       close vdac
[*]       close lvds
[*]       close mipi
[*]       close CVBS
[*]       close hdrx
[*]       close ddr
```

Configuration note:

If you need to use standby during the bootloader phase, you must first select the standby driver in the BT's menuconfig

- Build command: make kernel-rebuild all

Power consumption issue

- Standby mode will turn off most IPs, lower the CPU to 24M, and reduce DDR power consumption. Then wait in the cache to be woken up; when wake conditions are met, trigger watchdog reset, reboot the chip, achieving wake-up.
- If DDR supports GPIO-controlled power-off, DDR can be completely powered down. In the standby node, ddr-gpio = <1 0>, this parameter means that after entering standby, pin 1 is pulled low.
- If there are high-power circuits around the chip, you can control all peripheral circuits with pin 1, achieving the lowest CPU power in standby and turning off peripheral circuits to reach standby's minimum power consumption.

Wakeup

Currently standby supports three wakeup methods:

- IR wakeup
  - When using infrared wakeup, add parameter in the standby node (ir = <28>)
    - This parameter means that after entering standby, if an IR key press is detected and the decoded scancode is 28, infrared wakeup is triggered.
- ADC key wakeup
  - When using ADC key wakeup, add parameter in the standby node (adc = <1 0 850>)
    - This parameter means that after entering standby, if the voltage of ADC channel 1 is between 0 mV and 850 mV, ADC key wakeup is triggered.
- GPIO key wakeup
  - When using GPIO key wakeup, add parameter in the standby node (gpio = <2 0>)
    - This parameter means that after entering standby, if pin 2 is low, GPIO wakeup is triggered.

Debug

If you encounter issues with standby, enable this option, press the power key, and send the log to the technical support engineer. Menuconfig configuration

```
> Components
    > kernel
        > Drivers
            — standby driver
                --- standby driver
                Mode (Debug output hichip scancode) --->
```

Note: This configuration can only be used for debugging. After debugging is complete, it must be set to (Mode (Enable close ip) --->) to achieve low power consumption; otherwise, power savings will not be possible.

- Determine whether standby has been entered

  After entering standby via the application, pressing any infrared key will cause the serial port to print the scancode corresponding to the infrared key; if the power key is pressed, it will reboot.

  This method is also used to obtain the key's scancode.

Common issues

- Linux standby cannot wake up

    Check whether the ADC and IR drivers are configured in the AVP DTS and whether ADC and IR drivers are enabled in menuconfig.

- When powering on into standby, the IR cannot wake up.

    Check whether the IR driver is configured in the bootloader.

- Infrared cannot wake up.

    The IR parameters in the standby node of DTS are set incorrectly; this position is not a key value but the scancode obtained after driver decoding.

- ADC cannot wake up.

    ADC parameters in the standby node of DTS are set incorrectly. The maximum and minimum voltages must not be reversed.